Student Performance

Preprocessing, Statistics and Visualization

ENDTERM REPORT

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Objectives and analytical questions

01 How are final grades distributed?

02 What changes when missing values are imputed or removed?

03 Which original factors have the strongest observed association with G3?

04 How do group comparisons and visualizations qualify the findings?

Dataset and provenance

Property	Selected data
Subject	Portuguese language
Records / columns	649 / 33
Schools	GP: 423 · MS: 226
Original missing cells	0
Outcome	G3, final grade on a 0–20 scale
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Two Portuguese schools; findings do not directly represent TDTU students.

Inspect before transforming

```
raw = pd.read_csv(path, sep=';')
raw.head(5)
raw.tail(5)
raw.shape
raw.dtypes
raw.info()
```

649 × 33

Numeric, ordinal and categorical fields

Preserve raw data

Recorded zero grades remain observations; they are not reclassified as missing.

A controlled missingness experiment

649

complete original records

32

masked absences values
4.93% of records
Seed 42

Two treatments

Mean imputation

Drop missing rows

The missingness exercise is separate from the primary analysis of original measurements.

Imputation and deletion change the data

Variant	Rows	Missing	Mean absences	Sample SD
Original	649	0	3.6595	4.6408
Simulated missing	649	32	3.6240	4.6092
Mean imputation	649	0	3.6240	4.4940
Drop missing	617	0	3.6240	4.6092

Imputation preserves rows and concentrates replacement values at the mean.

A single seeded simulation does not establish a universally best treatment.

Python structures and regex

```
grades = raw['G3'].tolist()
selected = [g for g in grades if g >= 10]

grade_cols = [c for c in raw.columns
              if re.fullmatch(r'G[123]', c)]
```

Filtered list: 549 grades

Full matching validates actual schema names without inventing personal identifiers.

Indexing and a two-condition filter

```
df.loc[:9]
df.iloc[:, selected_positions]

df.loc[(df.studytime >= 2) &
       (df.absences <= 5)]
```

334

51.46% of records

Both conditions hold

The study-time variable is an ordered interval code, not a continuous hour count.

Three engineered features

Feature	Definition	Purpose
prior_grade_mean	(G1 + G2) / 2	Earlier-grade summary
grade_progress	G2 − G1	Assessment difference
performance_band	0–9; 10–14; 15–20	Outcome description

DataFrame.insert() places prior_grade_mean immediately before G3.

All derived features are excluded from the original-factor association ranking.

Grouped statistics and transformation

School	n	G3 mean	G3 median	G3 SD
GP	423	12.577	13	2.626
MS	226	10.650	11	3.834

Within-school score

(G3 − school minimum) / (school maximum − school minimum)

Constant groups receive 0; the scale describes relative position within each school.

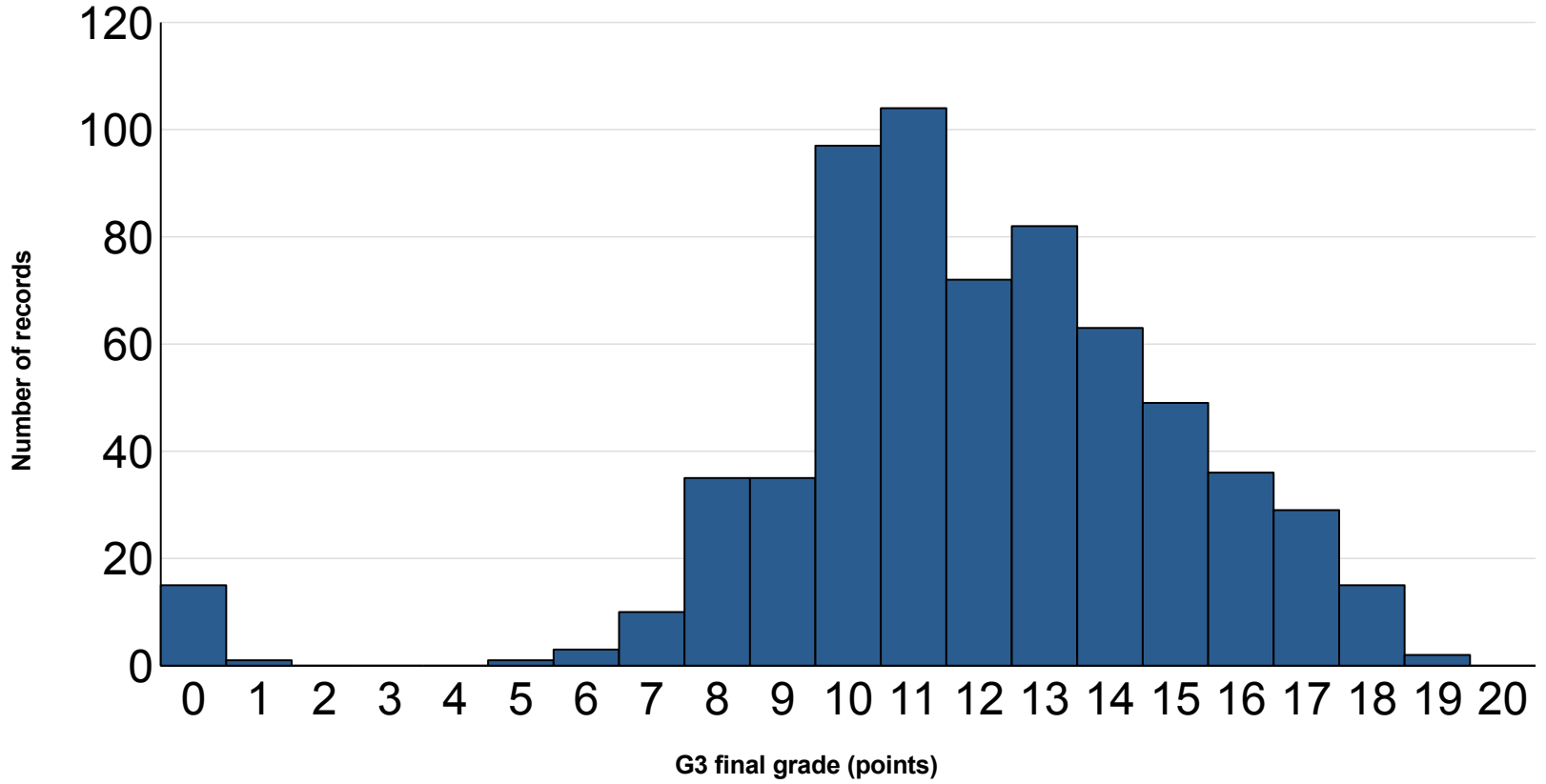
Association ranking: two distinct families

Prior grade: G2, Spearman ρ = 0.9445

Contextual numerical/ordinal factor	Spearman ρ	n
failures	-0.4484	649
Medu	0.2839	649
studytime	0.2747	649
Fedu	0.2346	649
Dalc	-0.2084	649

The ranking is unadjusted association among examined original numerical/ordinal factors.

Final-grade distribution



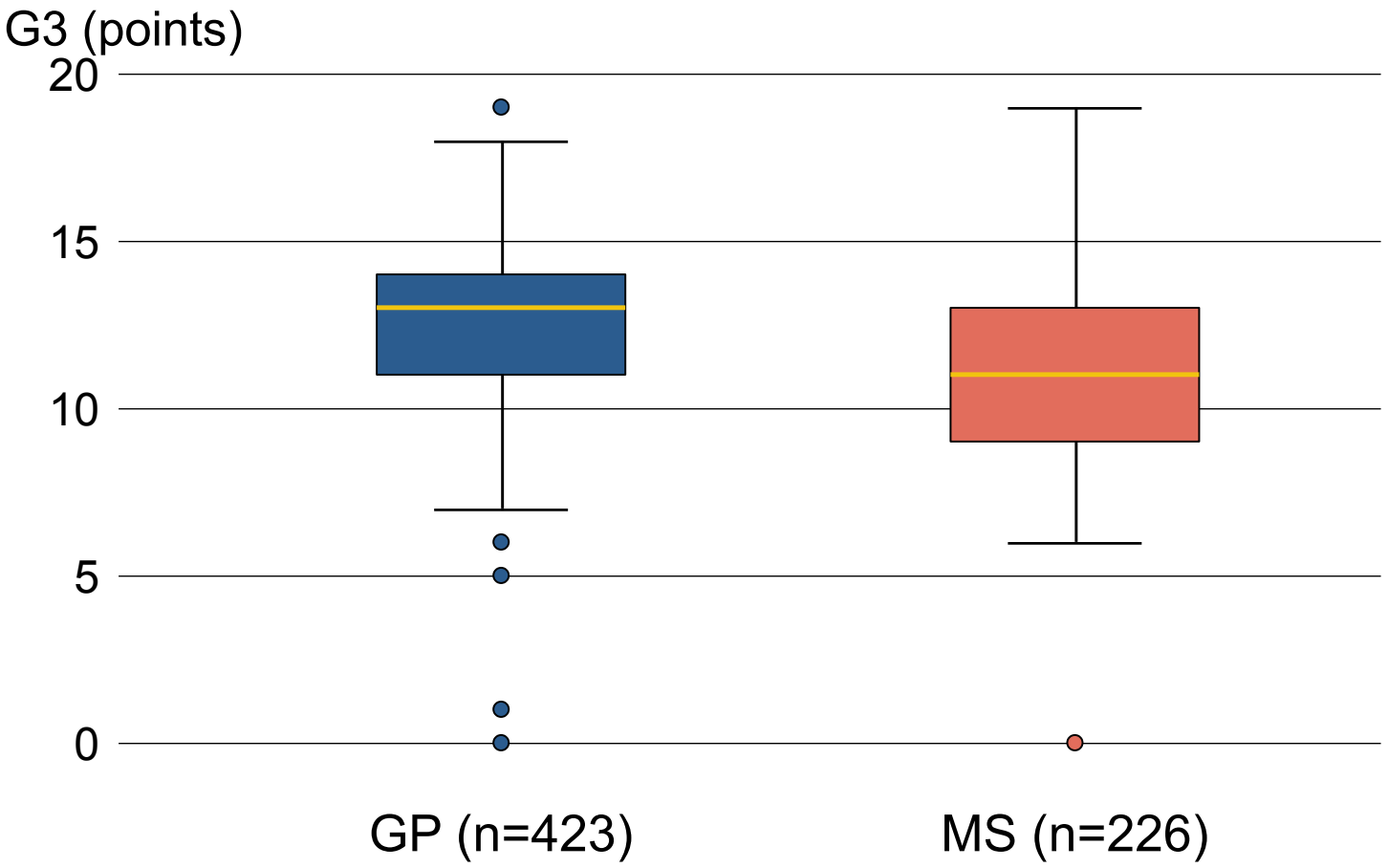
Mean 11.91

Median 12
Sample SD 3.23

n = 649

One-point-wide bins retain the integer-score distribution.

Final grades by school



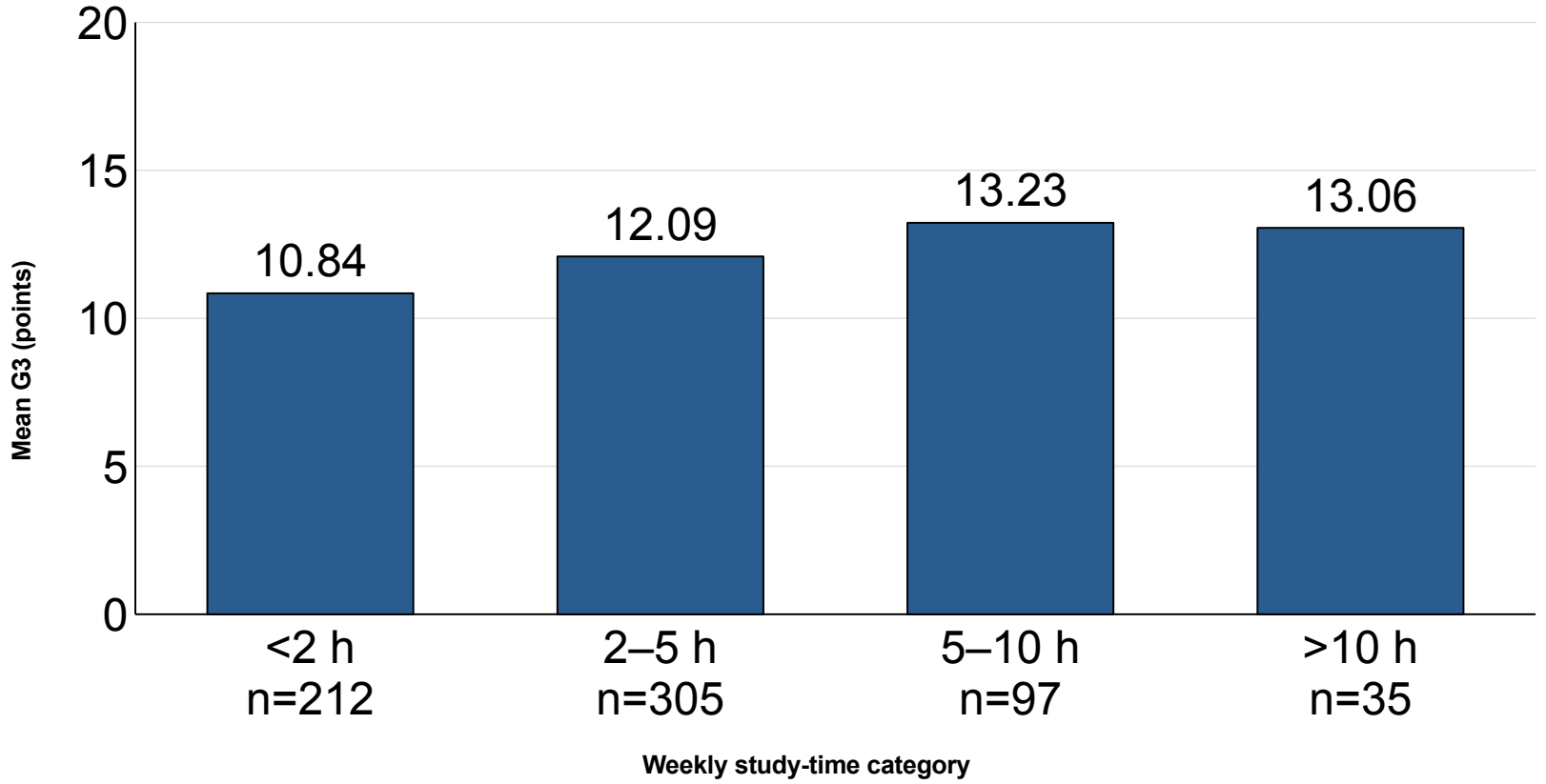
Medians: 13 / 11

GP has a higher observed centre.

MS has greater dispersion.

Descriptive group differences do not identify a causal school effect.

Final grade by study-time category



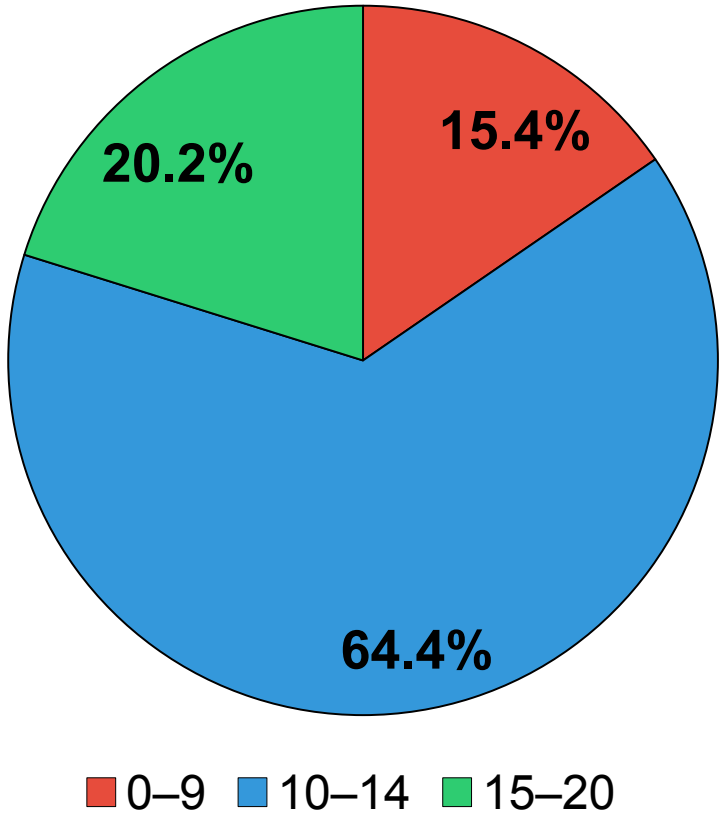
Unequal groups

Category codes represent intervals.

Counts accompany means.

The positive association remains observational and potentially confounded.

Performance-band proportions



Largest band
10–14

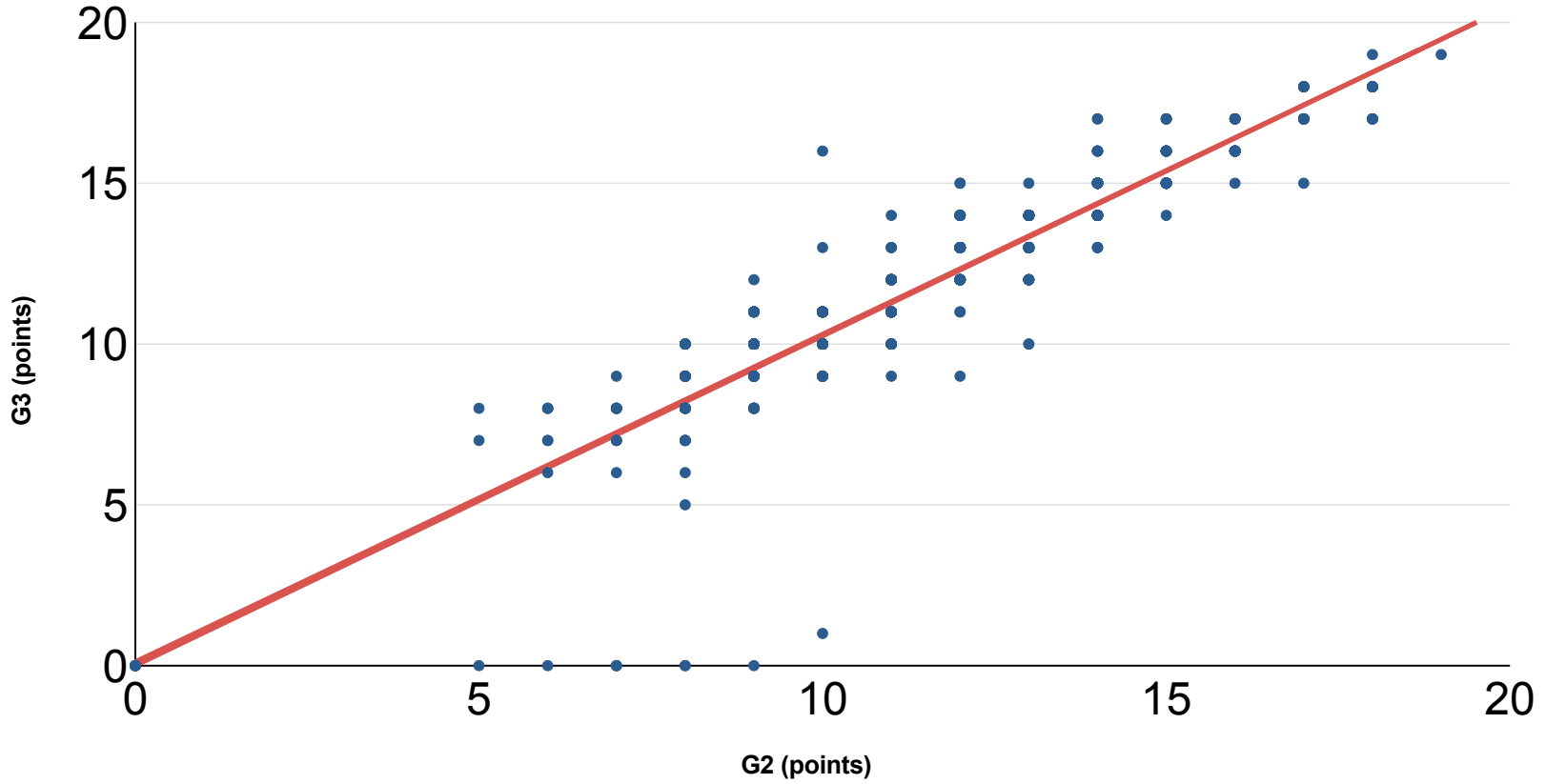
0–9: 100
15.41%

10–14: 418
64.41%

15–20: 131
20.18%

The bands are analyst-defined descriptions, not an official grading classification.

Earlier and final grades



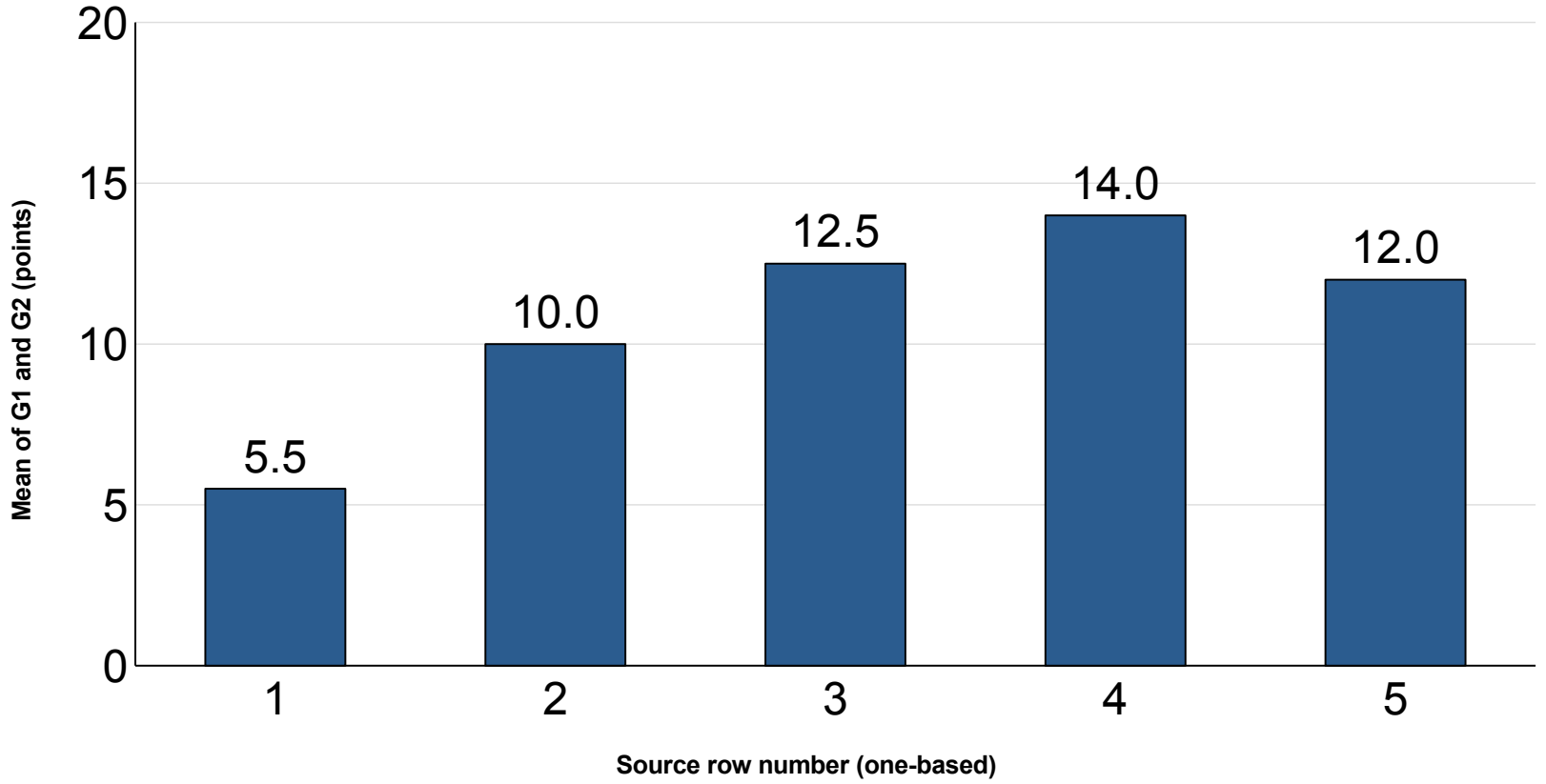
ρ = 0.9445

Pearson r = 0.9185

Related assessments at different periods

The fitted line describes the observations; it is not a validated forecasting model.

Matplotlib core from Python structures



List + dictionary

Only Matplotlib plots the notebook demonstration.

Five records illustrate mechanics.

This small example is not a random sample or a population trend.

Comprehensive report and extreme records

Summary item	Count	Share
Analysis records	649	100%
Analysis columns	37	33 original + 4 derived
G3 band 0–9	100	15.41%
G3 band 10–14	418	64.41%
G3 band 15–20	131	20.18%

Top/bottom: exactly ten rows; original source order resolves ties.

Complete rows and all columns are preserved in the notebook and CSV exports.

Code demonstration

01 Show the same 32-cell mask and both treatment branches.

02 Inspect insert(), grouped statistics and transform().

03 Call comprehensive_report(df) and inspect complete top/bottom rows.

04 Use the executed notebook if live execution fails.

Conclusions and limitations

Observed findings

Final grade mean: 11.91

G2 association: $\rho = 0.9445$

Strongest examined contextual rank association: failures

Interpretation limits

Two schools and one subject

Ordinal coding and unequal groups

Unadjusted observational associations

One MCAR simulation

References

Cortez (2008). Student Performance. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Pandas documentation https://pandas.pydata.org/docs/
Seaborn API https://seaborn.pydata.org/api.html
Matplotlib documentation https://matplotlib.org/stable/
NumPy documentation https://numpy.org/doc/

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Questions

Missing data, association, group comparison and reproducibility

Appendix: highest ten records

Source index	School	G1	G2	G3	Band
338	GP	18	19	19	15–20
636	MS	17	18	19	15–20
113	GP	18	17	18	15–20
181	GP	17	17	18	15–20
185	GP	16	17	18	15–20
314	GP	16	17	18	15–20
327	GP	17	17	18	15–20
332	GP	18	18	18	15–20
337	GP	17	18	18	15–20
349	GP	18	18	18	15–20

Selected columns shown; the full 37-column rows are in the notebook and CSV.

Appendix: lowest ten records

Source index	School	G1	G2	G3	Band
163	GP	11	9	0	0–9
440	MS	7	0	0	0–9
519	MS	8	7	0	0–9
563	MS	7	0	0	0–9
567	MS	4	0	0	0–9
583	MS	8	6	0	0–9
586	MS	8	8	0	0–9
597	MS	9	0	0	0–9
603	MS	5	0	0	0–9
605	MS	5	0	0	0–9

Selected columns shown; the full 37-column rows are in the notebook and CSV.

Appendix: sensitivity and method checks

Variant	n	Absences SD	ρ with G3	Filter n
Original	649	4.6408	-0.1585	334
Mean imputation	649	4.4940	-0.1628	339
Drop missing	617	4.6092	-0.1671	319

Boundary checks: score bands, constant groups, zero-count categories and stable ties.

Actual failure codes are retained; the upper-code ambiguity is documented.